

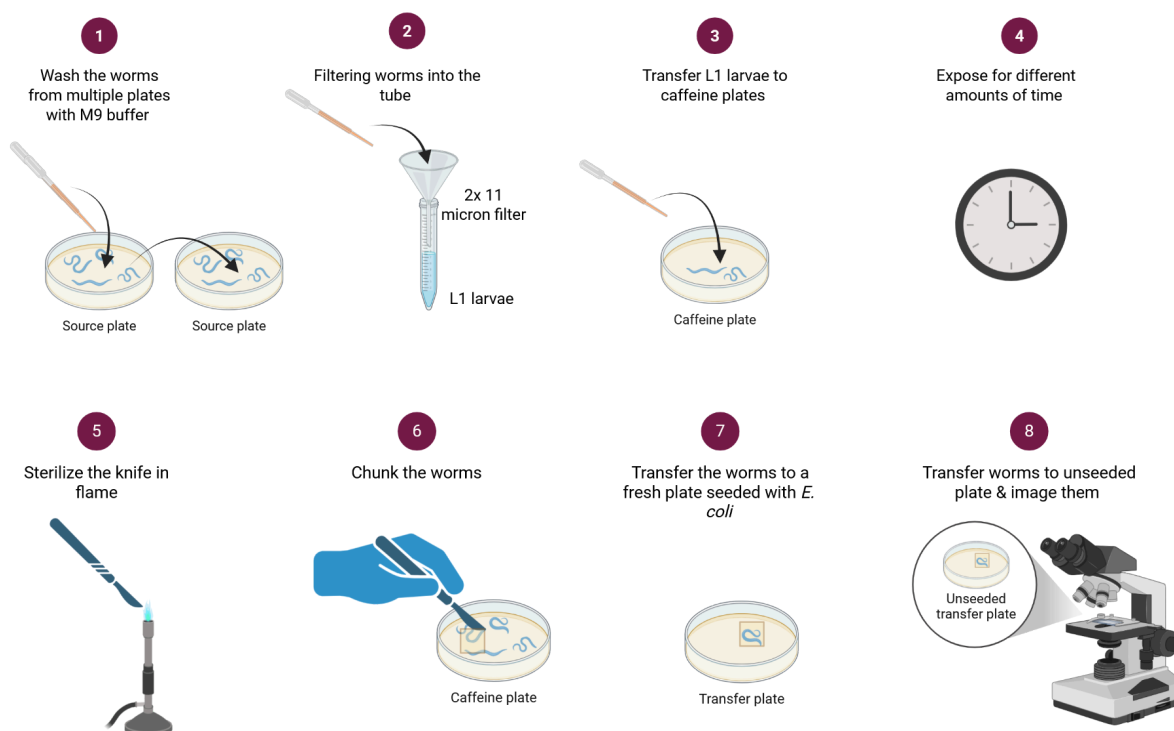
The Effect of Timing of Early-Life Caffeine Exposure on *C. elegans* morphology

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Number of rounds / repeats: 3

QBIO403: Developmental Biology



2.1 *C. elegans* strain maintenance

C. elegans (N2 wild type Bristol strain) was maintained on Nematode Growth Medium (NGM) seeded with an *E. coli* OP50 lawn throughout the experimental time frame. To maintain stocks, every 4-5 days, worms are transferred by cutting out a piece of agar with a sterilized knife and placing it on a fresh seeded plate. This is known as “chunking”.

2.2 *C. elegans* L1 synchronization procedure

Worms were washed from maintenance plates containing worms of all life stages with M9 buffer, and transferred onto a filtration apparatus with a plastic pipette. This filtration apparatus consisted of a Funnel with 2 20 µm filters, (Nylon net, Merck millipore), with a Falcon Tube (15ml) placed underneath. This allowed only L1 larvae to pass through.

After waiting 20 minutes for L1 worms to filter through and collect at the bottom, supernatant was discarded until approximately 2 mL of M9 buffer with the synchronized L1 population was left. The tube was inverted several times before pipetting one drop of the solution onto caffeine-containing NGM plates or control plates.

For the last round, 11 μ m filters were used to achieve a better synchronized population.

2.3 Caffeine Exposure from L1 to L4

For exposure to caffeine during development, NGM plates were also prepared with 5mM of caffeine (99.7 %, Thermo Scientific Chemicals) and seeded with *E. coli*.

The following caffeine exposure regimens were used, with 3 replicate plates for each condition:

Control	1h Exposure	4h Exposure	12h Exposure	20h Exposure	40h Exposure
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Worms were transferred from caffeine plates onto fresh NGM plates using the chunking method, corresponding to different exposure regimens. For the control and 40h exposure condition, worms were raised on the control NGM plates and caffeine NGM plates respectively for the entire 40h.

2.4 Imaging procedure

Round 1: Imaging was done through the removal of the agar from the plate with a knife to ensure direct physical access to the surface of the agar. Images were taken with an adaptive mobile microscope (APL-MS002BK) that was placed on a smartphone.

Round 2: Imaging was done using the Nikon SMZ745T microscope with an integrated camera and ThorCam (v. 3.7.0.6).

Round 3: Prior to imaging with the microscope, worms were first washed off seeded plates with M9 and pipetted onto an unseeded agar plate for imaging. The aim was to have evenly spread out worms that are not touching and not too close to the wall.

2.5 Data processing and analysis

Measurement of worm size was done using the WormSizer plugin (v. 1.2.5) in Fiji (v. 1.54p).

Measurement of velocity was done using TrackMate (v8.1.6). Plots were created with Python (v. 3.9.23).

Scan the QR-Code for more details (Fiji macros, parameter settings, and analysis scripts)!

